Statistics 652 – Homework 2

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Notation and a preliminary lemma

For a distribution function F on $\mathbb{R}$, let

$$Q_F(u) = \inf\{x : F(x) \geq u\}, \qquad 0 < u < 1,$$

and let Δ_x denote unit point mass at x. Define

$$F_{\varepsilon,x} = (1-\varepsilon)F + \varepsilon\Delta_x, \qquad 0 \leq \varepsilon < 1.$$

The influence function and gross-error sensitivity are

$$\mathrm{IF}(x;T,F) = \lim_{\varepsilon\downarrow 0} \frac{T(F_{\varepsilon,x}) - T(F)}{\varepsilon}, \qquad \gamma^*(T,F) = \sup_{x\in\mathbb{R}} |\,\mathrm{IF}(x;T,F)|.$$

All sampling statements concern independent identically distributed observations unless otherwise stated. An asymptotic variance is the variance of the root-n limiting distribution, not that variance divided by n.

Regular M-functional lemma. Let $t_0 = T(F)$ solve

$$\lambda_F(t_0) = 0, \qquad \lambda_F(t) = \mathbb{E}_F\psi(X,t).$$

Suppose λ_F is differentiable at t_0, with $A_F = \lambda_F'(t_0) \neq 0$. For a fixed contamination point x, assume $\psi(x,\cdot)$ is continuous at t_0 and that the selected roots $t_\varepsilon = T(F_{\varepsilon,x})$ satisfy $t_\varepsilon \to t_0$. Then

$$\mathrm{IF}(x;T,F) = -A_F^{-1}\psi(x,t_0).$$

Proof. The contaminated estimating equation is

$$(1-\varepsilon)\lambda_F(t_\varepsilon) + \varepsilon\psi(x,t_\varepsilon) = 0.$$

Put $d_\varepsilon = t_\varepsilon - t_0$. Differentiability gives $\lambda_F(t_\varepsilon) = (A_F + r_\varepsilon)d_\varepsilon$, where $r_\varepsilon \to 0$. Hence

$$\frac{d_\varepsilon}{\varepsilon} = -\frac{\psi(x,t_\varepsilon)}{(1-\varepsilon)(A_F + r_\varepsilon)} \longrightarrow -A_F^{-1}\psi(x,t_0).$$

□

For a smooth population equation, a nonzero parameter derivative and the implicit function theorem supply the required continuous local root branch. For a vector parameter, the same argument uses a nonsingular Jacobian. Differentiability of λ_F does not by itself license replacing λ_F' by $\mathbb{E}_F\partial_t\psi$; that interchange requires justification.

2.1. Recall that the mean is $T(F) = \mathbb{E}_F X$ and the median is $T(F) = F^{-1}(0.5)$.

(a) For continuous F, express the trimmed and Winsorized means as functionals $T(F)$.

(b) For absolutely continuous F, derive the median's influence function and gross-error sensitivity, and evaluate $\mathbb{E}_F[\mathrm{IF}(X; T, F)]$ and $\mathbb{E}_F[\mathrm{IF}(X; T, F)^2]$.

Solution:

Assumptions. Use symmetric trimming with $0 < \alpha < 1/2$. For the median derivative, assume F is absolutely continuous and differentiable at its median m, with $f(m) = F'(m) > 0$.

Proof.

(a) Let $a = Q_F(\alpha)$ and $b = Q_F(1-\alpha)$. Since $Q_F(U)$ has CDF F for $U \sim \mathrm{Uniform}(0,1)$, averaging the retained central probability mass gives

$$T_{\mathrm{tr},\alpha}(F) = \frac{1}{1-2\alpha} \int_\alpha^{1-\alpha} Q_F(u)\, du.$$

For continuous F, this equals

$$\frac{1}{1-2\alpha} \int_a^b x\, dF(x).$$

The Winsorizing map is $w_{a,b}(x) = \min\{b, \max\{a, x\}\}$. Its expectation is

$$T_{\mathrm{W},\alpha}(F) = \alpha a + \int_\alpha^{1-\alpha} Q_F(u)\, du + \alpha b.$$

No renormalization is needed because Winsorizing moves the tail mass to the cutoffs rather than deleting it. These quantile representations also define the functionals for distributions with atoms and require no tail moment assumptions.

(b) Write $m_\varepsilon = Q_{F_{\varepsilon,x}}(1/2)$. Because $F'(m) > 0$, for all sufficiently small $\delta > 0$,

$$F(m-\delta) < 1/2 < F(m+\delta).$$

The same strict inequalities hold under sufficiently small contamination. Therefore $m_\varepsilon \to m$. Fix $x \neq m$. The contaminating atom then stays on one side of a neighborhood of m, so

$$m_\varepsilon = Q_F(u_\varepsilon), \qquad u_\varepsilon = \frac{\frac{1}{2} - \varepsilon \mathbf{1}\{x < m\}}{1-\varepsilon}.$$

In particular,

$$u_\varepsilon - \frac{1}{2} = \varepsilon\left(\frac{1}{2} - \mathbf{1}\{x < m\}\right) + O(\varepsilon^2).$$

To differentiate the quantile, put $y = Q_F(u)$ as $u \to 1/2$. The strict inequalities around m imply $y \to m$, and continuity of F gives $F(y) = u$. Consequently,

$$\frac{Q_F(u) - m}{u - 1/2} = \frac{y - m}{F(y) - F(m)} \longrightarrow \frac{1}{f(m)}.$$

Factoring the contamination difference quotient now proves

$$\mathrm{IF}(x; T, F) = \frac{\frac{1}{2} - \mathbf{1}\{x < m\}}{f(m)} = \frac{\mathrm{sgn}(x - m)}{2f(m)}, \qquad x \neq m.$$

At $x = m$, the contaminated median remains m, giving influence zero. The median influence is therefore bounded by $1/(2f(m))$. Since $\Pr(X < m) = \Pr(X > m) = 1/2$, its positive and negative values have equal probabilities, and its square is constant almost surely.

Final answer.

$$\boxed{T_{\mathrm{tr},\alpha}(F) = \frac{\int_\alpha^{1-\alpha} Q_F(u)\,du}{1 - 2\alpha}}$$

$$\boxed{T_{\mathrm{W},\alpha}(F) = \alpha Q_F(\alpha) + \int_\alpha^{1-\alpha} Q_F(u)\,du + \alpha Q_F(1-\alpha)}$$

$$\boxed{\mathrm{IF}(x; \mathrm{Med}, F) = \frac{\mathrm{sgn}(x - m)}{2f(m)}, \quad \mathrm{sgn}(0) = 0; \qquad \gamma^* = \frac{1}{2f(m)}}$$

$$\boxed{\mathbb{E}_F\,\mathrm{IF}(X; T, F) = 0, \qquad \mathbb{E}_F[\mathrm{IF}(X; T, F)^2] = \frac{1}{4f(m)^2}.}$$

2.2. (a) Show that the mode is not qualitatively robust.

(b) Show that the trimmed and Winsorized means can be expressed as L-statistics: find $a_1, \ldots, a_n$ such that $T_n = \sum_{i=1}^n a_i x_{(i)}$.

Solution:

Assumptions. For (a), interpret a mode estimator as a sequence consistently estimating the unique population mode at the distributions used below. Qualitative

robustness refers to weak perturbations of the sampling distribution. For (b), use integer trimming of k observations from each tail, with $0 \le 2k < n$.

Proof.

(a) Let $F = N(0,1)$ with density ϕ and mode zero. Fix $a \neq 0$, and put

$$K(u) = \frac{3}{4}(1-u^2)\mathbf{1}\{|u| \le 1\}, \qquad h_\varepsilon = \varepsilon^2,$$

$$g_\varepsilon(t) = (1-\varepsilon)\phi(t) + \frac{\varepsilon}{h_\varepsilon} K\left(\frac{t-a}{h_\varepsilon}\right).$$

For its distribution G_ε,

$$d_{\mathrm{TV}}(G_\varepsilon, F) \le \varepsilon, \quad \text{so} \quad G_\varepsilon \Rightarrow F.$$

Nevertheless, $g_\varepsilon(a) = (1-\varepsilon)\phi(a) + 3/(4\varepsilon)$ tends to infinity, whereas outside $[a-h_\varepsilon, a+h_\varepsilon]$ the density is bounded by $\sup \phi$. All global modes lie inside that interval for small ε. There,

$$g_\varepsilon''(t) = (1-\varepsilon)\phi''(t) - \frac{3}{2}\varepsilon^{-5} < 0$$

for sufficiently small ε. Hence the mode is unique and

$$m_\varepsilon = T_{\mathrm{mode}}(G_\varepsilon) \to a \neq 0.$$

Thus the mode functional is not weakly continuous.

To establish non-robustness of an estimator sequence, let T_n be consistent at F and every fixed G_ε above. Write d_π for Prokhorov distance. Qualitative robustness at F would imply: for each $\zeta > 0$ there is $\delta > 0$ such that

$$d_\pi(G, F) < \delta \implies d_\pi(\mathcal{L}_G(T_n), \mathcal{L}_F(T_n)) < \zeta$$

uniformly in n (the argument also applies to the asymptotic version). Choose $0 < \zeta < \min\{|a|, 1\}/2$, then fix small $\varepsilon < \delta$ with $\min\{|m_\varepsilon|, 1\} > \zeta$. Since $d_\pi(G_\varepsilon, F) \le \varepsilon$, robustness gives the above bound. Consistency and passage to $n \to \infty$ imply

$$d_\pi(\Delta_{m_\varepsilon}, \Delta_0) \le \zeta.$$

But $d_\pi(\Delta_s, \Delta_t) = \min\{|s-t|, 1\}$, a contradiction.

Such consistent estimators exist for these distributions. For example, use a Gaussian-kernel density estimator with bandwidth $b_n = n^{-1/4}$ and a measurable

global maximizer as the mode estimator. Integration by parts and the empirical-CDF bound give

$$\sup_t |\widehat{f}_n(t) - \mathbb{E}\widehat{f}_n(t)| \le \frac{\|\phi'\|_1}{b_n} \sup_y |F_n(y) - F(y)| = O_p(n^{-1/4}).$$

For each fixed density above, bounded uniform continuity gives uniformly vanishing bias. Each density tends to zero at infinity and has a unique mode; hence that maximum is separated from the supremum outside any fixed neighborhood. Uniform density consistency implies mode consistency. Therefore the failure is one of qualitative robustness of consistent mode estimation, not merely a discontinuity statement without an estimator interpretation.

(b) For the trimmed mean,

$$T_{\mathrm{tr},n} = \frac{1}{n-2k} \sum_{i=k+1}^{n-k} x_{(i)}.$$

For the Winsorized mean,

$$T_{\mathrm{W},n} = \frac{1}{n} \left[k x_{(k+1)} + \sum_{i=k+1}^{n-k} x_{(i)} + k x_{(n-k)} \right].$$

Collecting the coefficient of each order statistic gives the desired L-statistic representations. The formula below also handles the case $k+1 = n-k$.

Final answer. The mode functional is not weakly continuous, and any mode estimator consistent at the displayed distributions is not qualitatively robust at $N(0,1)$. The L-statistic coefficients are

$$\boxed{a_i^{\mathrm{tr}} = \frac{\mathbf{1}\{k+1 \le i \le n-k\}}{n-2k}}$$

and

$$\boxed{a_i^{\mathrm{W}} = \frac{1}{n} \left[\mathbf{1}\{k+1 \le i \le n-k\} + k\mathbf{1}\{i = k+1\} + k\mathbf{1}\{i = n-k\} \right].}$$

For an exact empirical quantile-integral definition rather than integer trimming, let $I_i = ((i-1)/n, i/n]$ and let λ be length. Then the corresponding coefficients are

$$a_i^{\mathrm{tr},\alpha} = \frac{\lambda(I_i \cap [\alpha, 1-\alpha])}{1-2\alpha},$$

$$a_i^{\mathrm{W},\alpha} = \lambda(I_i \cap [\alpha, 1-\alpha]) + \alpha\mathbf{1}\{i = \lceil n\alpha \rceil\} + \alpha\mathbf{1}\{i = \lceil n(1-\alpha) \rceil\}.$$

These exact-quantile conventions need not coincide with integer trimming at boundary ranks.

2.3. For a location M-estimate, show (i) $\rho(x,t) = \rho(x-t)$ with $\rho(-r) = \rho(r)$, and (ii) $\psi(x,t) = \psi(x-t)$ with $\psi(-r) = -\psi(r)$.

Solution:

Assumptions. Use a translation-invariant, reflection-symmetric loss:

$$\rho(x+a, t+a) = \rho(x,t), \qquad \rho(-x,-t) = \rho(x,t).$$

Assume differentiability when an ordinary score is used. Reflection symmetry is an additional assumption: it does not follow from location equivariance alone. For example, $\rho_\tau(r) = r(\tau - \mathbf{1}\{r < 0\})$ is a location loss but is not even for $\tau \neq 1/2$.

Proof. Taking $a = -t$ gives $\rho(x,t) = \rho(x-t, 0)$. Define $\rho_0(r) = \rho(r,0)$. Then

$$\rho(x,t) = \rho_0(x-t), \qquad \rho_0(-r) = \rho(-r,0) = \rho(r,0) = \rho_0(r).$$

Let $\psi_0 = \rho_0'$. Differentiating the evenness identity yields $-\psi_0(-r) = \psi_0(r)$, so ψ_0 is odd and $\psi_0(0) = 0$. The minimizing first-order equation is

$$\sum_i \psi_0(x_i - t) = 0.$$

If $\psi(x,t)$ is defined as $\partial_t \rho(x,t)$, it equals $-\psi_0(x-t)$; this convention gives the same roots and oddness. For convex nonsmooth losses, replace the derivative equation by $0 \in \sum_i \partial\rho_0(x_i - t)$.

If F is symmetric about θ and the score is integrable, oddness also implies $\mathbb{E}_F \psi_0(X - \theta) = 0$. Consequently the functional is Fisher consistent whenever that is the unique selected root.

Final answer. Under translation invariance and reflection symmetry,

$$\boxed{\rho(x,t) = \rho_0(x-t), \quad \rho_0(-r) = \rho_0(r), \qquad \psi_0(-r) = -\psi_0(r).}$$

2.4. Derive the asymptotic relative efficiency of the mean to the median at (i) $X \sim \mu + \sigma t_\nu$ and (ii) $X \sim (1-\eta)N(\mu, \sigma^2) + \eta N(\mu, 9\sigma^2)$.

Solution:

Assumptions and convention. Define $\mathrm{ARE}(A, B) = V_B/V_A$, where V is root-n asymptotic variance. Thus an ARE exceeding one favors A. For a common mean and unique median μ, finite variance v, and differentiability at μ with $f(\mu) > 0$, the CLT and 2.9 give

$$V_{\text{mean}} = v, \qquad V_{\text{median}} = \frac{1}{4f(\mu)^2}, \qquad \mathrm{ARE}(\bar{X}, \widehat{m}) = \frac{1}{4vf(\mu)^2}.$$

Proof.

(a) For $\nu > 2$,

$$v = \sigma^2 \frac{\nu}{\nu - 2}, \qquad f_X(\mu) = \frac{c_\nu}{\sigma}, \qquad c_\nu = \frac{\Gamma((\nu+1)/2)}{\sqrt{\nu\pi}\,\Gamma(\nu/2)}.$$

Substitution gives

$$\mathrm{ARE}(\bar{X}, \widehat{m}) = \frac{\nu - 2}{4\nu c_\nu^2} = \frac{\pi(\nu-2)}{4} \left[\frac{\Gamma(\nu/2)}{\Gamma((\nu+1)/2)}\right]^2.$$

The gamma-ratio asymptotic $\Gamma(z + 1/2)/\Gamma(z) \sim z^{1/2}$ gives the limit $\pi/2$. For example, the ARE is $\pi^2/16$ at $\nu = 3$, $8/9$ at $\nu = 4$, and approximately 1.04093 at $\nu = 5$. The crossover is $\nu \approx 4.67879$.

For $0 < \nu \le 2$, the ordinary ratio of finite root-n variances is not defined for the sample mean. The median remains root-n normal for every $\nu > 0$. Student tails are regularly varying with index $-\nu$: for $0 < \nu < 2$, the generalized CLT gives

$$n^{1-1/\nu}(\bar{X} - \mu) \xrightarrow{d} S_\nu,$$

where S_ν is a nondegenerate symmetric stable law with the appropriate scale. Thus the mean is slower than root-n for $1 < \nu < 2$, does not concentrate for $\nu = 1$, and has increasing spread for $0 < \nu < 1$. At $\nu = 2$, its concentration rate is of order $\sqrt{n/\log n}$, again slower than root-n. Accordingly its efficiency relative to the median is zero under the extended rate-based convention. For $\nu \le 1$, μ denotes location, not a finite expectation.

(b) The component means agree, so

$$v = (1 - \eta)\sigma^2 + \eta(9\sigma^2) = \sigma^2(1 + 8\eta).$$

Symmetry gives median μ, and

$$f_X(\mu) = \frac{1-\eta}{\sigma\sqrt{2\pi}} + \frac{\eta}{3\sigma\sqrt{2\pi}} = \frac{1 - \frac{2}{3}\eta}{\sigma\sqrt{2\pi}}.$$

Substitution yields the expression below. Its denominator apart from the factor two is $D(\eta) = (1+8\eta)(1-\frac{2}{3}\eta)^2$, with

$$D'(\eta) = \frac{4}{3}(1-\tfrac{2}{3}\eta)(5-12\eta).$$

Thus the ARE decreases until $\eta = 5/12$ and then increases; it is not monotone on $[0,1]$. Solving ARE $= 1$ gives approximately 0.100592 and 0.814126. At $\eta = 0$ and 1, the distribution is a single normal and the ARE is $\pi/2$.

Final answer.

$$\boxed{\mathrm{ARE}(\bar{X}, \widehat{m}) = \frac{\pi(\nu-2)}{4}\left[\frac{\Gamma(\nu/2)}{\Gamma((\nu+1)/2)}\right]^2, \quad \nu > 2.}$$

For $\nu \le 2$, finite root-n variance ARE is unavailable; the extended rate-based efficiency of the mean relative to the median is zero.

$$\boxed{\mathrm{ARE}(\bar{X}, \widehat{m}) = \frac{\pi}{2(1+8\eta)(1-\frac{2}{3}\eta)^2}, \quad 0 \le \eta \le 1.}$$

The median is more efficient for $0.100592\ldots < \eta < 0.814126\ldots$.

2.5. Let $\widehat{\theta}_n$ minimize $\sum_i \rho(x_i;\theta)$, with ρ the negative log-likelihood of a regular one-parameter family. Show directly, without MLE asymptotic theory, that at $F = F_\theta$, $\mathbb{E}_F[\mathrm{IF}(X;T,F)^2] = 1/I(\theta;X)$.

Solution:

Assumptions. Assume an identifiable regular family with common support, differentiation under the integral justified up to second order, and $0 < I(\theta;X) < \infty$. Assume the population score equation is continuously differentiable near the model parameter and under small contamination. The functional follows the local minimizer branch through θ; if defined by global minimization, assume its minimizer stays on this branch. Expected log-likelihood differences below are assumed well-defined.

Proof. Put $s_t(x) = \partial_t \log f_t(x)$ and $\psi(x,t) = \partial_t \rho(x;t) = -s_t(x)$. Fisher consistency follows from

$$\mathbb{E}_\theta \rho(X;t) - \mathbb{E}_\theta \rho(X;\theta) = D_{\mathrm{KL}}(F_\theta \| F_t) \ge 0,$$

with equality only at $t = \theta$ by identifiability. Thus $T(F_\theta) = \theta$. Differentiating $\int f_\theta = 1$ once and twice gives

$$\mathbb{E}_\theta s_\theta(X) = 0, \qquad \mathbb{E}_\theta[\partial_\theta s_\theta(X) + s_\theta(X)^2] = 0.$$

Consequently,

$$A_{F_\theta} = -\mathbb{E}_\theta \partial_\theta s_\theta(X) = \mathbb{E}_\theta s_\theta(X)^2 = I(\theta; X) > 0.$$

The implicit function theorem supplies a differentiable local root branch, and the preliminary lemma gives

$$\mathrm{IF}(x; T, F_\theta) = I(\theta; X)^{-1} s_\theta(x).$$

Squaring and integrating proves the result directly.

Final answer.

$$\boxed{\mathrm{IF}(x; T, F_\theta) = \frac{s_\theta(x)}{I(\theta; X)}, \qquad \mathbb{E}_\theta[\mathrm{IF}(X; T, F_\theta)^2] = \frac{1}{I(\theta; X)}.}$$

For a vector parameter and nonsingular information matrix, the corresponding identity is $\mathbb{E}_\theta[\mathrm{IF}\,\mathrm{IF}^\top] = I(\theta)^{-1}$.

2.6. For a distribution with a finite second moment, express its variance σ_F^2 as a functional, find its influence function and gross-error sensitivity, sketch the influence function, evaluate its first two moments, and derive the sample variance's asymptotic variance by the CLT.

Solution:

Assumptions and notation. Let $\mu = \mathbb{E}_F X$, $v = \sigma_F^2$, and $\mu_4 = \mathbb{E}_F[(X-\mu)^4]$. The functional is defined on $\mathcal{F}_2 = \{F : \int x^2\, dF(x) < \infty\}$. For the CLT assume $\mu_4 < \infty$. Here σ_F^2 is the variance; the wording "standard deviation" and centering by σ in 2.6 are interpreted consistently with the sample variance as σ_F^2.

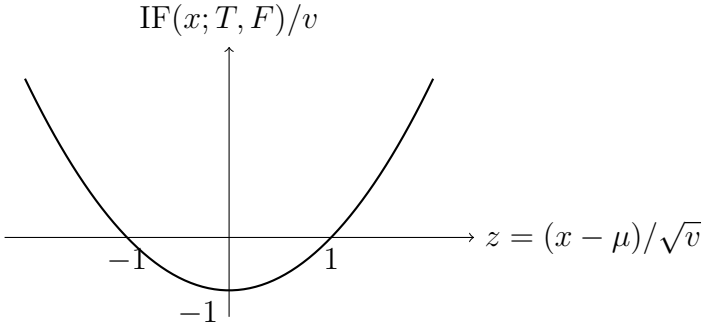
Proof. The functional is

$$T(F) = \int x^2\, dF(x) - \left(\int x\, dF(x)\right)^2.$$

The contaminated mean is $\mu + \varepsilon(x - \mu)$, and the mixture variance identity gives exactly

$$T(F_{\varepsilon,x}) = (1-\varepsilon)v + \varepsilon(1-\varepsilon)(x-\mu)^2.$$

Differentiating yields $\mathrm{IF}(x;T,F) = (x-\mu)^2 - v$. This diverges as $|x| \to \infty$. Its graph is an upward-opening parabola with vertex $(\mu, -v)$ and, for $v > 0$, zeros $\mu \pm \sqrt{v}$.



The displayed standardized sketch assumes $v > 0$; when $v = 0$, the unscaled graph is $(x-\mu)^2$. Taking moments,

$$\mathbb{E}_F \,\mathrm{IF}(X;T,F) = \mathbb{E}_F[(X-\mu)^2] - v = 0,$$

$$\mathbb{E}_F[\mathrm{IF}(X;T,F)^2] = \mathbb{E}_F[((X-\mu)^2 - v)^2] = \mu_4 - v^2.$$

The last quantity is infinite if $\mu_4 = \infty$.

For $s_n^2 = n^{-1}\sum_i (X_i - \bar{X})^2$, the exact identity

$$s_n^2 = \frac{1}{n}\sum_i (X_i-\mu)^2 - (\bar{X}-\mu)^2$$

implies

$$\sqrt{n}(s_n^2 - v) = \frac{1}{\sqrt{n}}\sum_i [(X_i-\mu)^2 - v] - \sqrt{n}(\bar{X}-\mu)^2.$$

By the mean CLT, $\bar{X} - \mu = O_p(n^{-1/2})$, so the final term is $o_p(1)$. The CLT for $(X_i-\mu)^2$ and Slutsky's theorem give

$$\sqrt{n}(s_n^2 - v) \xrightarrow{d} N(0, \mu_4 - v^2).$$

A zero limiting variance is allowed as a degenerate normal limit.

Final answer.

$$\boxed{T(F) = \mathbb{E}_F X^2 - (\mathbb{E}_F X)^2, \quad \mathrm{IF}(x;T,F) = (x-\mu)^2 - v, \quad \gamma^* = \infty.}$$

$$\boxed{\mathbb{E}_F \,\mathrm{IF} = 0, \qquad \mathbb{E}_F \,\mathrm{IF}^2 = \mu_4 - v^2 = \mathrm{AVar}(s_n^2).}$$

For normal F, this asymptotic variance is $2\sigma^4$.

2.7. Express the standard deviation σ as a functional, find and sketch its influence function, and determine whether the sample standard deviation's asymptotic variance equals $\mathbb{E}_F[\mathrm{IF}^2]$.

Solution:

Assumptions. Let $F \in \mathcal{F}_2$ and $\sigma > 0$. For asymptotic normality, assume $\mu_4 = \mathbb{E}_F[(X-\mu)^4] < \infty$.

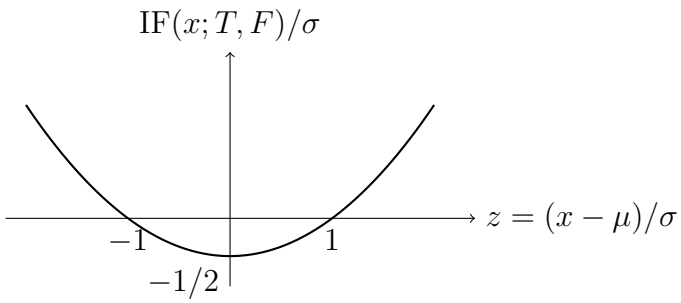
Proof. The standard deviation functional is the square root of the variance:

$$T(F) = \left[\int x^2\, dF(x) - \left(\int x\, dF(x)\right)^2\right]^{1/2}.$$

Applying the ordinary chain rule to the contamination derivative from 2.6 gives

$$\mathrm{IF}(x; T, F) = \frac{(x-\mu)^2 - \sigma^2}{2\sigma}.$$

This is an upward-opening parabola with vertex $(\mu, -\sigma/2)$ and zeros $\mu \pm \sigma$; it has infinite gross-error sensitivity.



Direct calculation gives

$$\mathbb{E}_F\,\mathrm{IF} = 0, \qquad \mathbb{E}_F\,\mathrm{IF}^2 = \frac{\mu_4 - \sigma^4}{4\sigma^2}.$$

Applying the delta method to 2.6 with $g(v) = \sqrt{v}$ yields

$$\sqrt{n}(s_n - \sigma) \xrightarrow{d} N\left(0, \frac{\mu_4 - \sigma^4}{4\sigma^2}\right).$$

At $\sigma = 0$, $F = \Delta_\mu$, and exactly

$$T(F_{\varepsilon,x}) = \sqrt{\varepsilon(1-\varepsilon)}\,|x - \mu|.$$

For $x \neq \mu$, the contamination difference quotient is infinite in the limit, so there is no finite regular influence function there.

Final answer. For $\sigma > 0$,

$$\boxed{\mathrm{IF}(x; T, F) = \frac{(x-\mu)^2 - \sigma^2}{2\sigma}, \qquad \mathrm{AVar}(s_n) = \mathbb{E}_F\,\mathrm{IF}^2 = \frac{\mu_4 - \sigma^4}{4\sigma^2}.}$$

For normal F, the asymptotic variance is $\sigma^2/2$.

2.8. Derive the density of $X = \mu + \sigma RZ$, where $\mu \in \mathbb{R}$, $\sigma > 0$, $Z \sim N(0,1)$, $R > 0$ has density f_R, and R, Z are independent.

Solution:

Assumptions. Only positivity of R, the stated independence, and existence of the mixing density are needed; no moments of R are required.

Proof. Conditioning gives $X \mid R = r \sim N(\mu, \sigma^2 r^2)$ and

$$f_{X|R}(x \mid r) = \frac{1}{\sigma r\sqrt{2\pi}} \exp\left[-\frac{(x-\mu)^2}{2\sigma^2 r^2}\right].$$

Integrating this conditional density against f_R gives the result. Tonelli's theorem verifies normalization:

$$\int_{\mathbb{R}} f_X(x)\,dx = \int_0^\infty f_R(r)\left[\int_{\mathbb{R}} f_{X|R}(x \mid r)\,dx\right] dr = 1.$$

The density is symmetric about μ. Its mixture version at the center has value $\mathbb{E}[R^{-1}]/(\sigma\sqrt{2\pi})$, possibly infinite.

Final answer.

$$\boxed{f_X(x) = \int_0^\infty \frac{\exp[-(x-\mu)^2/(2\sigma^2 r^2)]}{\sigma r\sqrt{2\pi}} f_R(r)\,dr.}$$

For a general mixing law H_R, replace $f_R(r)\,dr$ by $dH_R(r)$; this includes discrete scale mixtures.

2.9. (Optional.) Prove consistency and asymptotic normality of the sample median: if μ is the population median and $f(\mu) > 0$, establish the limiting distribution of $\sqrt{n}(\widehat{\mu}_n - \mu)$.

Solution:

Assumptions. Assume F is continuous near its unique median μ and is differentiable at μ with derivative $f(\mu) > 0$. Let $k_n = \lceil n/2 \rceil$ and initially set $\widehat{\mu}_n = X_{(k_n)}$. The centered limiting normal has mean zero; the mean μ printed in the assignment's limiting law is corrected accordingly.

Proof of consistency. For every $\delta > 0$,

$$F(\mu - \delta) < 1/2 < F(\mu + \delta).$$

If $\widehat{\mu}_n > \mu + \delta$, then $F_n(\mu + \delta) < k_n/n$. By the law of large numbers and $k_n/n \to 1/2$, this event has probability tending to zero. Likewise $\widehat{\mu}_n < \mu - \delta$ implies $F_n(\mu - \delta) \geq k_n/n$, again with probability tending to zero. Therefore $\widehat{\mu}_n \xrightarrow{p} \mu$.

Proof of asymptotic normality. Fix $t \in \mathbb{R}$ and set

$$y_n = \mu + t/\sqrt{n}, \qquad p_n = F(y_n) = \frac{1}{2} + \frac{f(\mu)t}{\sqrt{n}} + o(n^{-1/2}).$$

The count $B_n = \sum_i \mathbf{1}\{X_i \leq y_n\}$ is $\mathrm{Bin}(n, p_n)$. Since $p_n \to 1/2$, the standardized Bernoulli triangular array satisfies Lindeberg's condition, giving

$$Z_n = \frac{B_n - np_n}{\sqrt{np_n(1-p_n)}} \xrightarrow{d} N(0,1).$$

The order-statistic event satisfies

$$\Pr(\widehat{\mu}_n \leq y_n) = \Pr(B_n \geq k_n) = 1 - \Pr\left(Z_n \leq \frac{k_n - 1 - np_n}{\sqrt{np_n(1-p_n)}}\right).$$

The threshold converges to $-2f(\mu)t$. Convergence to the continuous normal CDF permits this moving threshold, so the last expression tends to $\Phi(2f(\mu)t)$.

For even $n = 2k$, the same argument applies to ranks k and $k+1$. For $M_n = (X_{(k)} + X_{(k+1)})/2$,

$$\Pr(\sqrt{n}(X_{(k+1)} - \mu) \leq t) \leq \Pr(\sqrt{n}(M_n - \mu) \leq t) \leq \Pr(\sqrt{n}(X_{(k)} - \mu) \leq t).$$

Both outer terms have the same limit, proving the result for the usual averaged median without an order-statistic spacing assumption.

Final answer.

$$\boxed{\widehat{\mu}_n \xrightarrow{p} \mu, \qquad \sqrt{n}(\widehat{\mu}_n - \mu) \xrightarrow{d} N\left(0, \frac{1}{4f(\mu)^2}\right).}$$

2.10. Show that if X_1, X_2 are independent Cauchy(μ, σ^2) variables, then $(X_1 + X_2)/2$ has the same distribution.

Solution:

Assumptions and parameterization. Let $s > 0$ denote the Cauchy scale under the assignment's convention. Its density is $s/\{\pi[(x-\mu)^2+s^2]\}$. The second parameter in the problem is a scale convention, not a finite variance.

Proof. The characteristic function is $\phi_X(t) = \exp(i\mu t - s|t|)$. Independence gives

$$\phi_{(X_1+X_2)/2}(t) = \phi_{X_1}(t/2)\phi_{X_2}(t/2) = \exp(i\mu t - s|t|) = \phi_X(t).$$

Uniqueness of characteristic functions proves the claim.

Final answer.

$$\boxed{(X_1 + X_2)/2 \stackrel{d}{=} X_1 \sim \text{Cauchy}(\mu, \sigma^2).}$$

The same calculation gives $\bar{X}_n \stackrel{d}{=} X_1$ for every n.

2.11. A standardized symmetric stable variable with index $0 < \alpha \leq 2$ has characteristic function $\phi_X(t) = \exp(-|t|^\alpha)$. For i.i.d. copies X_i and $Y_i = \mu+\sigma X_i$, find the characteristic function of $\bar{Y}_n = n^{-1}\sum_i Y_i$ and describe how its spread depends on n.

Solution:

Assumptions. Let $\sigma > 0$ and use the stated characteristic-function standardization. Spread is measured by scale or fixed interquantile ranges, not by variance when it is infinite.

Proof. Independence gives

$$\begin{aligned}\phi_{\bar{Y}_n}(t) &= e^{i\mu t}\prod_{i=1}^{n}\exp\left[-\left|\frac{\sigma t}{n}\right|^\alpha\right] \\ &= \exp\left[i\mu t - \sigma^\alpha n^{1-\alpha}|t|^\alpha\right].\end{aligned}$$

This is the characteristic function of $\mu+\sigma n^{1/\alpha-1}X$. Therefore its interquantile ranges are multiplied by $n^{1/\alpha-1}$.

Final answer.

$$\boxed{\phi_{\bar{Y}_n}(t) = \exp\left[i\mu t - \sigma^\alpha n^{1-\alpha}|t|^\alpha\right], \qquad \bar{Y}_n \overset{d}{=} \mu + \sigma n^{1/\alpha - 1} X.}$$

Spread decreases for $\alpha > 1$, stays constant for $\alpha = 1$, and increases for $0 < \alpha < 1$. At $\alpha = 2$, the standard law is $N(0, 2)$ and the scale decreases as $n^{-1/2}$. For $\alpha < 2$, the variance is infinite; for $\alpha \leq 1$, μ is location rather than a finite mean.

2.12. Prove that $\mathbb{E}_F[\mathrm{IF}(X; T, F)] = 0$ whenever T is an M-functional.

Solution:

Assumptions. Let $T(F) = t_0$ solve $\lambda_F(t_0) = \mathbb{E}_F\psi(X, t_0) = 0$, with $\mathbb{E}_F|\psi(X, t_0)| < \infty$. Assume $A_F = \lambda_F'(t_0) \neq 0$, continuous selected contaminated roots, and continuity of $\psi(x, \cdot)$ at t_0 for F-almost every x. These are regularity conditions for the influence function to exist; singular or nonunique nonregular functionals need not possess one.

Proof. The preliminary lemma gives, for F-almost every x,

$$\mathrm{IF}(x; T, F) = -A_F^{-1}\psi(x, t_0).$$

The integrability assumption makes this influence function integrable. Taking expectations and using the population estimating equation,

$$\mathbb{E}_F \mathrm{IF}(X; T, F) = -A_F^{-1}\mathbb{E}_F\psi(X, t_0) = 0.$$

The argument only requires differentiability of the population expectation. For example, with the median score $\psi(x, t) = \mathrm{sgn}(x - t)$ and continuous F,

$$\lambda_F(t) = 1 - 2F(t), \qquad A_F = -2f(m).$$

The pointwise score derivative is zero off its jump, so taking its expectation would incorrectly miss this nonzero population derivative. For Huber's Lipschitz score, bounded difference quotients instead justify dominated differentiation, yielding

$$A_F = -\Pr_F(|X - t_0| < c)$$

when there is no mass at the two kinks. The vector case follows identically with a nonsingular Jacobian.

Final answer. For every regular M-functional satisfying the stated conditions,

$$\boxed{\mathbb{E}_F[\mathrm{IF}(X; T, F)] = 0.}$$

2.13. Show that Tukey's M-estimate of location does not correspond to a maximum likelihood estimate under a symmetric location model.

Solution:

Assumptions and criterion. For $c > 0$, use Tukey's bisquare loss

$$\rho_c(r) = \begin{cases} \dfrac{c^2}{6}\left[1 - \left(1 - \dfrac{r^2}{c^2}\right)^3\right], & |r| \le c, \\ \dfrac{c^2}{6}, & |r| > c, \end{cases}$$

with score $\psi_c(r) = r(1-r^2/c^2)^2\mathbf{1}\{|r| \le c\}$. A likelihood representation of the defining loss or score is considered first. To rule out even an alternative regular likelihood having the same estimator for all samples, assume its symmetric location density g is positive and continuously differentiable.

Proof. If the loss were a negative log-density up to a positive scale and an additive constant, then

$$g(r) = C\exp[-a\rho_c(r)], \qquad C > 0,\ a > 0.$$

For $|r| > c$, this is the positive constant $Ce^{-ac^2/6}$. Hence it cannot integrate to one. The same conclusion follows from matching likelihood scores: $-(\log g)' = a\psi_c$ integrates to the same improper expression. This obstruction to a loss/score likelihood representation does not require smoothness assumptions on a candidate density beyond the proposed representation itself.

There is also a direct estimator-level contradiction for any positive continuously differentiable symmetric density, even without requiring proportional scores. Consider the three observations $(0, 0, r)$ with $r > 2c$, and write $C_c = c^2/6$. The Tukey objective is

$$L(t) = 2\rho_c(t) + \rho_c(r - t).$$

If $|t| < c$, then $|r - t| > c$, so $L(t) = 2\rho_c(t) + C_c$, uniquely minimized there at $t = 0$. If $|t| \ge c$, then $L(t) \ge 2C_c > C_c = L(0)$. Thus its unique global minimizer is zero.

If this estimator were also the MLE under g for every sample, zero would maximize $g(-t)^2 g(r - t)$. The derivative of its log-likelihood at zero must vanish:

$$-2\frac{g'(0)}{g(0)} - \frac{g'(r)}{g(r)} = 0.$$

Symmetry and differentiability imply $g'(0) = 0$, hence $g'(r) = 0$ for every $r > 2c$. Thus g is a positive constant on $(2c, \infty)$, contradicting integrability.

Final answer.

Tukey's bisquare loss/score has no proper location likelihood representation.

Moreover, no positive continuously differentiable symmetric location density has the same location MLE as Tukey's global M-estimator for all samples. The bounded loss, equivalently the score's exact tail cutoff, is the obstruction. Huber's linearly growing tail loss does not have this obstruction: its negative exponential is integrable.

2.14. For the location-scale double-exponential distribution, find the MLE, show that the scale estimate has an unbounded influence function, and find the finite-sample breakdown point of the scale estimate.

Solution:

Assumptions and parameterization. Use

$$f(x;\mu,b)=\frac{1}{2b}\exp\left(-\frac{|x-\mu|}{b}\right),\quad b>0.$$

Its standard deviation is $\sqrt{2}b$. For influence calculations, assume a finite first moment, absolute continuity, and $f(m)>0$ at the median. For breakdown, take $n\geq 2$ and an original sample with positive fitted scale. For even n, select the averaged median.

Proof of the MLE. The log-likelihood is

$$\ell(\mu,b)=-n\log(2b)-\frac{1}{b}\sum_i|x_i-\mu|.$$

For fixed μ, its maximizing scale is $\widehat{b}(\mu)=n^{-1}\sum_i|x_i-\mu|$ when the sum is positive. Profiling reduces the location problem to minimizing the sum of absolute deviations, whose minimizers are the sample medians. The minimum sum is the same for every median in the even-sample median interval. If all data coincide, the likelihood diverges as $b\downarrow 0$ and no positive-scale MLE exists; extend the fitted scale continuously by zero for breakdown calculations.

Proof of the influence function. Let $H(t)=\mathbb{E}_F|X-t|$, $m=m(F)$, and $B(F)=H(m)$. The bound

$$||X-t-h|-|X-t||\leq|h|$$

justifies dominated differentiation at points without an atom, so $H'(t) = 2F(t) - 1$ and $H'(m) = 0$. Problem 2.1 gives $m_\varepsilon - m = O(\varepsilon)$ for fixed x. Thus $H(m_\varepsilon) = H(m) + o(\varepsilon)$, and

$$B(F_{\varepsilon,x}) = (1-\varepsilon)H(m_\varepsilon) + \varepsilon|x - m_\varepsilon| = B(F) + \varepsilon(|x-m| - B(F)) + o(\varepsilon).$$

Therefore $\mathrm{IF}(x; B, F) = |x - m| - B(F)$. At the Laplace model, $m = \mu$ and $B(F) = b$, giving unbounded influence $|x - \mu| - b$. In contrast, the location influence is $b\,\mathrm{sgn}(x - \mu)$, which is bounded.

Proof of replacement breakdown. Replace one observation by $M \to \infty$, keeping an original observation x_j fixed. For every center t,

$$|M - t| + |x_j - t| \geq |M - x_j|.$$

Consequently,

$$\widehat{b}_M = \frac{1}{n} \min_t \sum_i |x'_i - t| \geq \frac{|M - x_j|}{n} \longrightarrow \infty.$$

One replacement therefore causes scale explosion. To include implosion, let r be the maximum multiplicity of an original value. Replacing $n - r$ observations by that value makes $\widehat{b} = 0$. With fewer replacements, at least $r + 1$ original observations remain, including two distinct values. If $d > 0$ is the smallest positive separation of distinct original values, then $\widehat{b}' \geq d/n$, preventing implosion. Thus the implosion fraction is $(n - r)/n$, and the explosion fraction is $1/n$. Their minimum is $1/n$.

Final answer.

$$\boxed{\widehat{\mu} = \mathrm{median}(x_1, \ldots, x_n), \qquad \widehat{b} = \frac{1}{n} \sum_i |x_i - \widehat{\mu}|.}$$

$$\boxed{\mathrm{IF}(x; B, F_{\mu,b}) = |x - \mu| - b, \quad \gamma^* = \infty, \qquad \epsilon_n^*(\widehat{b}; x) = \frac{1}{n}.}$$

The asymptotic scale breakdown point is zero. For an all-distinct sample, the implosion-only fraction is $(n - 1)/n$. Rescaling the parameter to $\sigma = \sqrt{2}b$ multiplies its influence function by $\sqrt{2}$ and leaves its breakdown point unchanged.

2.15. Find the MLEs of β and σ^2 for $Y_i = X_i^\top \beta + \epsilon_i$, with independent $\epsilon_i \sim N(0, \sigma^2)$, using exponential-family theory.

Solution:

Assumptions. Let X be the fixed $n \times p$ design matrix, of full column rank, with $n > p$. The parameter space is $\mathbb{R}^p \times (0, \infty)$.

Proof. The joint Gaussian density can be written in canonical form using

$$\eta = \frac{\beta}{\sigma^2}, \qquad \tau = -\frac{1}{2\sigma^2} < 0 :$$

$$f(y; \eta, \tau) = \exp\{\eta^\top X^\top y + \tau y^\top y - A(\eta, \tau)\},$$

where

$$A(\eta, \tau) = \frac{n}{2} \log\left(\frac{\pi}{-\tau}\right) - \frac{\eta^\top X^\top X \eta}{4\tau}.$$

The natural space $\mathbb{R}^p \times (-\infty, 0)$ is open. This family is minimal: if $a^\top X^\top y + b\, y^\top y$ is constant for all y, its Hessian forces $b = 0$, and full column rank then forces $a = 0$. Thus A is strictly convex and the canonical log-likelihood strictly concave. An interior solution of the canonical moment equations is the unique maximizer. The parameter transformation is bijective.

For the canonical statistic $S(Y) = (X^\top Y, Y^\top Y)$, the exponential-family identity $\nabla A = \mathbb{E} S(Y)$ yields

$$X^\top y = \mathbb{E}[X^\top Y] = X^\top X \beta,$$

$$y^\top y = \mathbb{E}[Y^\top Y] = \beta^\top X^\top X \beta + n\sigma^2.$$

The first equation gives the least-squares coefficient, and the second gives the residual sum of squares divided by n:

$$y^\top y - \widehat{\beta}^\top X^\top X \widehat{\beta} = \|y - X\widehat{\beta}\|^2.$$

If the residual sum of squares is positive, the solution is interior and hence is the MLE. If it is zero, an exact fit has likelihood unbounded as $\sigma^2 \downarrow 0$.

Final answer. For $y \notin \operatorname{col}(X)$,

$$\boxed{\widehat{\beta} = (X^\top X)^{-1} X^\top y, \qquad \widehat{\sigma}^2 = \frac{1}{n}\|y - X\widehat{\beta}\|^2.}$$

For $y \in \operatorname{col}(X)$, no positive-variance MLE exists. The MLE uses divisor n, not the unbiased-estimation divisor $n - p$.

2.16. Verify the adaptively weighted least-squares representation of a regression M-estimate

of β.

Solution:

Assumptions. Minimize

$$L(\beta) = \sum_i \rho(u_i(\beta)), \qquad u_i(\beta) = \frac{y_i - x_i^\top \beta}{s}, \quad s > 0,$$

with scale held fixed. Let ρ be even and differentiable, $\psi = \rho'$, and suppose $\psi(u)/u$ has a finite continuous extension at zero. For a weighted least-squares minimization, assume nonnegative weights and a nonsingular weighted design.

Proof. The first-order condition is

$$\sum_i x_i \psi(u_i(\widehat{\beta})) = 0.$$

Define

$$w(u) = \begin{cases} \psi(u)/u, & u \neq 0, \\ \psi'(0), & u = 0, \end{cases} \qquad W(\beta) = \operatorname{diag}(w(u_1(\beta)), \ldots, w(u_n(\beta))).$$

Since $\psi(u) = w(u)u$, the estimating equation becomes

$$X^\top W(\widehat{\beta})(y - X\widehat{\beta}) = 0.$$

Solving these weighted normal equations gives the desired implicit representation. Freezing weights at the current iterate gives IRLS. For Huber and Tukey scores respectively,

$$w_c^{\mathrm{H}}(u) = \begin{cases} 1, & |u| \leq c, \\ c/|u|, & |u| > c, \end{cases} \qquad w_c^{\mathrm{T}}(u) = \left(1 - \frac{u^2}{c^2}\right)^2 \mathbf{1}\{|u| \leq c\}.$$

For these losses, w is even, nonnegative, and nonincreasing in $|u|$. Let $g(v) = \rho(\sqrt{v})$ for $v \geq 0$. Since $g'(v) = w(\sqrt{v})/2$ for $v > 0$, g is concave, and its tangent inequality implies

$$\rho(u) \leq \rho(u_0) + \frac{1}{2} w(u_0)(u^2 - u_0^2).$$

The extension at $u_0 = 0$ follows by continuity. Summing these inequalities produces a quadratic majorizer touching L at the current iterate. Exact WLS minimization therefore cannot increase L, provided scale is fixed. This does not guarantee convergence to a global minimizer for a nonconvex loss.

Final answer.

$$\boxed{\widehat{\beta} = [X^\top W(\widehat{\beta})X]^{-1}X^\top W(\widehat{\beta})y,}$$

with iterative update

$$\boxed{\beta^{(k+1)} = [X^\top W(\beta^{(k)})X]^{-1}X^\top W(\beta^{(k)})y.}$$

The weights depend on fitted residuals, so the representation is a fixed-point identity, not a closed-form solution. Zero weights may make the design singular. Residual weighting alone does not generally control high-leverage design points. If scale is estimated separately, it is held fixed during this argument; alternating scale updates require a separate scale equation and convergence analysis.